CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

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Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03, R07, R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

In row R03, rule Q05 was broken because the inspection Status is 'READY', but the Inspector_ID field was left blank. The source data contains an inspection entry missing its corresponding inspector identity. This prevents tracking who conducted the procedure and invalidates accountability for that specific record.

Next action: what to verify, who owns it, and how to retest

The next step is to request the Shift Supervisor to verify the missing inspector assignment against the original physical logbook or digital system entries. Once the true identity is confirmed, the value should be populated in the source file, and the analysis re-run to confirm that rule Q05 no longer flags this row.

Limit: why you cannot guess a correction or prove accuracy

We cannot guess or invent the missing values because making assumptions compromises data integrity and could lead to false records. Furthermore, a clean spreadsheet with no rule violations only proves logical consistency within the file; it cannot prove that the recorded machine data matches the actual events on the factory floor.

Professional connection: one framework idea and how you used it

I utilized the concept of Completeness from the Data Quality Framework to address the missing data points in critical tracking fields. By explicitly flagging empty cells in identifiers like Inspector_ID and Machine_ID rather than guessing them, we guarantee that the report accurately reflects data gaps that require operational verification.

Refresh test: before, changed value, after, and restored result

Before the test, row R01 shows an Inspection_Minutes value of 20 and does not flag rule Q04. When temporarily changed to 0, rule Q04 successfully flags row R01 as out of range. After restoring the value back to 20 and clicking 'Refresh All', the error flag disappears, proving the automated Power Query audit pipeline functions correctly.

Export only Report: two landscape pages, page break before row 26. Review both pages.